



Islamic University of Technology
Department of Computer Science and Engineering

Lab 5: Shortest Path (Continued)

CSE 4404: Algorithms Lab(SWE)
Summer 2023-24

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Task A. Speedrun Strats

Time Limit: 2 seconds | *Memory Limit: 512 MB*

Zaid is a speedrunner of a popular dungeon-crawler game. He has spent months studying every room, every passage, and every glitch in the game.

The game has n rooms connected by m one-way passages. Each passage has a time delta — positive if the passage is slow (a long cutscene, a difficult platforming section), and negative if Zaid has discovered an out-of-bounds glitch that skips part of the level and saves time.

His viewers keep spamming the chat: “bro what’s the fastest route from room a to room b ??”

Help Zaid answer q such chat queries without pausing the run.

Input Format

The first line contains three integers n , m , and q ($1 \leq n \leq 300$, $1 \leq m \leq 3000$, $1 \leq q \leq 100$) — the number of rooms, passages, and queries.

Each of the next m lines contains three integers a , b , and c ($1 \leq a, b \leq n$, $-10^9 \leq c \leq 10^9$) — indicating a one-way passage from room a to room b with time delta c .

Each of the next q lines contains two integers a and b ($1 \leq a, b \leq n$) — a query asking for the minimum total time to travel from room a to room b .

It is guaranteed that the input contains no negative cycles.

Output Format

For each query, print the minimum total time on its own line.

If room b is unreachable from room a , print INF.

Examples

Sample Input	Sample Output
4 5 4	2
1 2 3	4
2 3 -1	1
1 3 5	INF
3 4 2	
1 4 10	
1 3	
1 4	
2 4	
4 1	

Notes

In the sample, there are 4 rooms and 5 passages:

- Query $1 \rightarrow 3$: the direct passage takes 5 seconds, but the route $1 \rightarrow 2 \rightarrow 3$ takes $3 + (-1) = 2$ seconds — passage $2 \rightarrow 3$ has a glitch that saves a second.
- Query $1 \rightarrow 4$: the fastest route is $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$ with total time $3 + (-1) + 2 = 4$.
- Query $2 \rightarrow 4$: route $2 \rightarrow 3 \rightarrow 4$ takes $(-1) + 2 = 1$ second.
- Query $4 \rightarrow 1$: room 1 is unreachable from room 4, so the answer is **INF**.

Task B. The Arbitrage Machine

Time Limit: 2 seconds | Memory Limit: 512 MB

Rahim is a currency trader with access to n different currencies and m one-way exchange rates between them. A one-way exchange rate from currency a to currency b with rate r means that 1 unit of currency a can be converted into r units of currency b .

Rahim suspects there may be an **arbitrage opportunity** — a sequence of exchanges starting and ending at the same currency such that the final amount is strictly greater than the initial amount.

For example, if 1 unit of currency A gives 2 units of B, 1 unit of B gives 2 units of C, and 1 unit of C gives 0.4 units of A, then starting with 1 unit of A yields $1 \times 2 \times 2 \times 0.4 = 1.6$ units of A — a clear profit.

Help Rahim determine whether any arbitrage opportunity exists.

Input Format

The first line contains two integers n and m ($1 \leq n \leq 2500$, $1 \leq m \leq 5000$) — the number of currencies and exchange rates.

Each of the next m lines contains two integers a, b and a real number r ($1 \leq a, b \leq n$, $10^{-3} \leq r \leq 10^3$) — indicating that 1 unit of currency a can be exchanged for r units of currency b . The value r is given with at most 6 decimal places.

Output Format

Print YES if an arbitrage opportunity exists. Otherwise, print NO.

Examples

Sample Input	Sample Output
3 3 1 2 2.0 2 3 2.0 3 1 0.3	YES
3 3 1 2 0.5 2 3 2.0 3 1 0.9	NO

Notes

In the first sample, the exchange sequence $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$ gives $1 \times 2.0 \times 2.0 \times 0.3 = 1.2 > 1$, so an arbitrage exists.

In the second sample, the only cycle $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$ gives $1 \times 0.5 \times 2.0 \times 0.9 = 0.9 < 1$, so no arbitrage exists.

Hint: Taking the negative logarithm of each exchange rate transforms this into a negative cycle detection problem on a weighted directed graph.

Task C. The Signal Leak

Time Limit: 2 seconds | Memory Limit: 512 MB

You are given a directed weighted graph of n nodes and m edges. Edge weights can be negative. Starting from source node s , find the minimum cost to reach every other node.

Some paths may pass through a **negative cycle** — a cycle whose total weight is negative. Any node reachable via such a cycle has an unbounded negative cost ($-\infty$).

Input Format

The first line contains three integers n , m , and s ($1 \leq n \leq 2500$, $1 \leq m \leq 5000$, $1 \leq s \leq n$) — the number of nodes, edges, and the source node.

Each of the next m lines contains three integers a , b , and c ($1 \leq a, b \leq n$, $-10^9 \leq c \leq 10^9$) — a directed edge from node a to node b with weight c .

Output Format

Print n lines. The i -th line contains the minimum cost from s to node i :

- If $i = s$, print 0.
- If node i is reachable via a negative cycle, print $-\text{INF}$.
- If node i is unreachable from s , print INF .
- Otherwise, print the minimum cost.

Examples

Sample Input	Sample Output
6 5 1	0
1 2 1	$-\text{INF}$
2 3 2	$-\text{INF}$
3 2 -5	$-\text{INF}$
3 4 1	8
1 5 8	INF

Notes

In the sample, starting from node 1:

- Edges $2 \rightarrow 3 \rightarrow 2$ form a negative cycle with total weight $2 + (-5) = -3 < 0$. Nodes 2 and 3 are in the cycle.
- Node 4 is downstream of the cycle (reachable from node 3), so it is also $-\text{INF}$.
- Node 5 is only reachable via $1 \rightarrow 5$ with cost 8, unaffected by any cycle.
- Node 6 has no incoming edges so it is unreachable from node 1: INF .

Marks Distribution

Task	Marks
Task A	30%
Task B	30%
Task C	40%

Practice Problems

Problem	Links
Signal Hill	DMOJ: https://dmoj.ca/problem/dmpg15s4 Vjudge: https://vjudge.net/problem/DMOJ-dmpg15s4
Roads in Berland	Codeforces: https://codeforces.com/contest/25/problem/C Vjudge: https://vjudge.net/problem/CodeForces-25C
High Score	CSES: https://cses.fi/problemset/task/1673 Vjudge: https://vjudge.net/problem/CSES-1673
Cycle Finding	CSES: https://cses.fi/problemset/task/1197 Vjudge: https://vjudge.net/problem/CSES-1197
Edge Deletion	AtCoder: https://atcoder.jp/contests/abc243/tasks/abc243_e Vjudge: https://vjudge.net/problem/AtCoder-abc243_e
Flight Discount	CSES: https://cses.fi/problemset/task/1195 Vjudge: https://vjudge.net/problem/CSES-1195
Greg and Graph	Codeforces: https://codeforces.com/problemset/problem/295/B Vjudge: https://vjudge.net/problem/CodeForces-295B